Name:Viswapriya S

Reg.no:212225230311

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## Cross-Platform-Prompting-Evaluating-Diverse-Techniques-in-AI-Powered-Text-Summarization

### Introduction:

**Cross-Platform Prompting: Evaluating Diverse Techniques in AI-Powered Text Summarization** is a study that explores how different prompt engineering methods perform across multiple Artificial Intelligence platforms for generating accurate and meaningful text summaries. With the rapid growth of Generative AI and Large Language Models (LLMs), text summarization has become an important application in education, business, healthcare, research, and communication systems.

The main objective of this topic is to analyze how various prompting techniques — such as straightforward prompts, contextual prompts, chain-of-thought prompting, and role-based prompting — influence the quality, clarity, and efficiency of summaries produced by different AI tools. Since AI platforms like OpenAI ChatGPT, Google Gemini, and Microsoft Copilot may respond differently to the same prompt, cross-platform evaluation helps identify the most effective prompting strategies.

This study also focuses on important summarization factors such as accuracy, coherence, brevity, contextual understanding, and response time. By comparing outputs from different AI systems, users can understand which prompting techniques provide better results for academic content, articles, reports, and long documents.

Overall, cross-platform prompting improves the effectiveness of AI-powered text summarization and helps researchers, students, and developers design better prompts for obtaining high-quality summaries from modern AI models.

## Objective of the Study:

The main aim of this study is to:

- Compare summaries generated by different AI platforms.
- Analyze how prompting techniques affect summarization quality.
- Identify which platform produces clearer and more accurate summaries.
- Evaluate the efficiency of AI models in understanding long passages.
- Understand how prompts can improve AI-generated outputs.

## AI Platforms Used

- OpenAI ChatGPT
- Google Gemini
- Microsoft Copilot

## Evaluation Criteria

1. Accuracy

Measures whether the summary correctly captures the important points from the original text.

**2. Coherence**

Checks whether the summary is logically organized and easy to understand.

**3. Simplicity**

Determines whether the language is simple enough for students and general users.

**4. Speed**

Measures how quickly the AI generates the summary.

**5. User Experience (UX)**

Evaluates the interface, readability, and ease of saving or using the generated output.

## Prompting Techniques Used

### 1. Zero-Shot Prompting:

**Definition**

**Zero-shot prompting means giving the AI a direct instruction without any examples.**

**Prompt Used**

**“Summarize the text without giving any examples.”**

**Explanation**

**In this method, the AI independently understands the content and generates a summary. The AI relies entirely on its training knowledge and language understanding capabilities.**

**Observations**

- **ChatGPT generated a concise and clear explanation.**

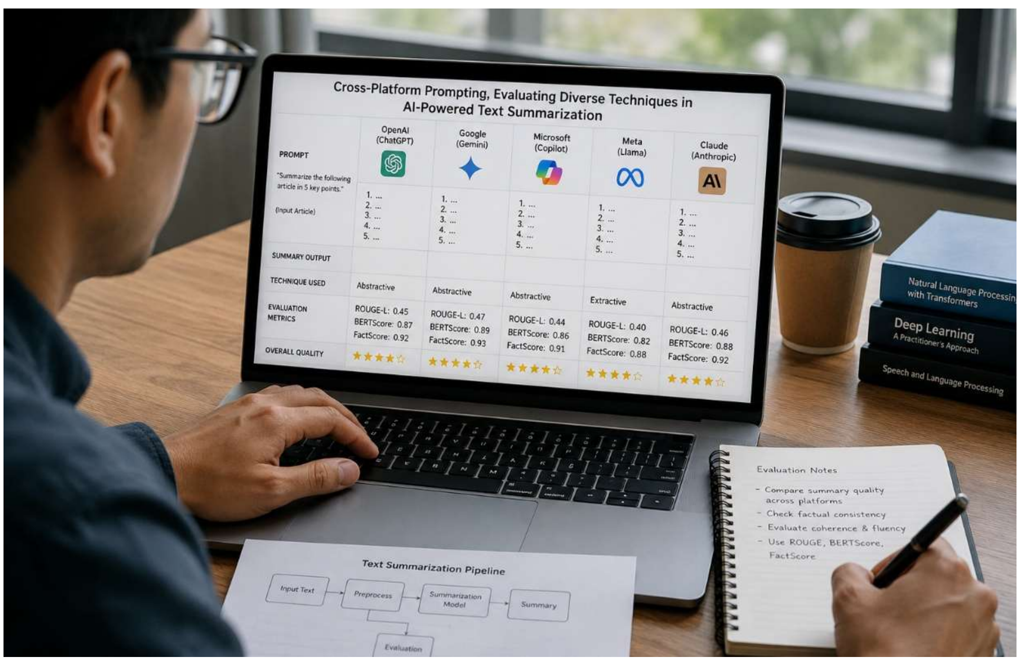
- **Gemini produced a structured point-wise summary.**
- **Copilot focused more on technical details and security aspects.**

**Advantages**

- **Fast and simple.**
- **Requires minimal prompt design.**
- **Useful for quick summarization.**

**Limitations**

- **Output quality may vary across platforms.**
- **Sometimes lacks depth or examples.**




## 2. Few-Shot Prompting:

Definition

Few-shot prompting provides instructions along with examples or guidance.

Prompt Used

"Summarize the text with giving two or three examples."

Explanation

Here, the AI generates summaries along with practical examples such as cryptocurrency, supply chain management, and decentralized finance.

Observations

- ChatGPT gave a balanced summary with application examples.
- Gemini provided highly organized bullet points.
- Copilot produced concise technical summaries with examples.

Advantages

- Improves clarity and understanding.
- Provides contextual examples.
- Better for educational purposes.

Limitations

- Longer responses.
- May include unnecessary details.

## 3. Chain-of-Thought Prompting

## Definition

Chain-of-thought prompting encourages AI to explain information step-by-step in detail.

Prompt Used

"Based on the above two summarization, give me the detailed summary of blockchain technology."

Explanation

This technique helps AI produce more detailed and logically connected summaries. The platforms explained blockchain concepts such as decentralization, cryptographic hashing, consensus mechanisms, and smart contracts in a structured manner.

Observations

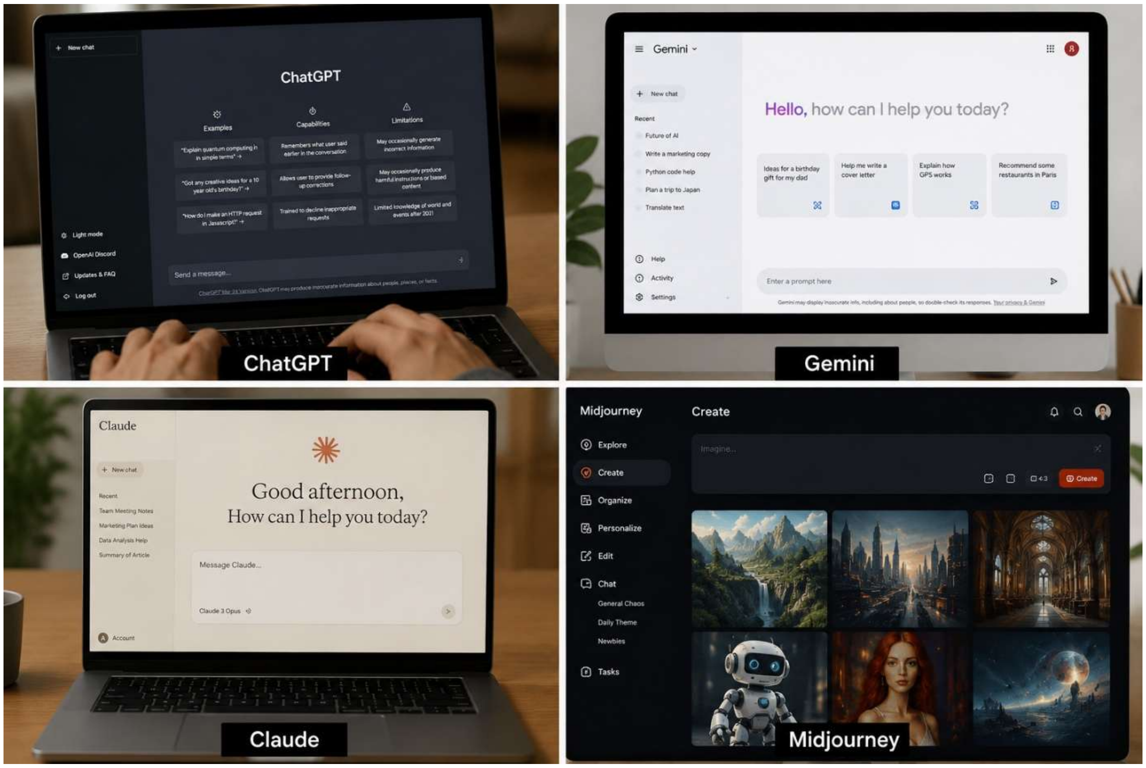
- ChatGPT generated detailed paragraph-based explanations.
- Gemini organized the explanation into sections and subtopics.
- Copilot used categorized headings and technical formatting.

Advantages

- Produces highly informative summaries.
- Improves reasoning and logical flow.
- Useful for research and academic work.

Limitations

- Takes more time to generate.
- Responses can become lengthy.




## 4.Role-Based Prompting

Definition

Role-based prompting asks the AI to respond from a specific professional perspective.

Prompt Used

“Consider you are a Researcher in the institution and now summarize the passage in that role perspective.”

Explanation

The AI generated summaries using a research-oriented and academic tone. The platforms focused on technical depth, analytical explanations, and research implications.

Observations

- ChatGPT emphasized distributed systems and research applications.

- Gemini produced an academic research brief format.
- Copilot highlighted conceptual rigor and technological impact.

Advantages

- Produces professional-quality summaries.
- Useful for academic and research presentations.
- Enhances contextual understanding.

Limitations

- May become too technical for beginners.
- Requires carefully designed prompts.

**Comparative Analysis of AI Platforms**

| Platform | Strengths | Weaknesses |
|---|---|---|
| OpenAI ChatGPT | Clear explanations, balanced summaries, easy readability | Sometimes lengthy |
| Google Gemini | Structured and organized responses | Occasionally too formal |
| Microsoft Copilot | Technical depth and concise output | Less conversational |

**Importance of Cross-Platform Prompting**

Cross-platform prompting is important because:

- Different AI models interpret prompts differently.
- It helps identify the best summarization strategy.
- Improves prompt engineering skills.

- Enhances AI-generated content quality.
- Supports research in Generative AI and Natural Language Processing (NLP).

## Conclusion

Cross-platform prompting demonstrates how different AI systems respond uniquely to the same summarization task. By evaluating techniques such as zero-shot, few-shot, chain-of-thought, and role-based prompting, users can understand which methods produce the most accurate, coherent, and meaningful summaries.

The study also highlights that prompt engineering plays a major role in improving AI-powered text summarization. Among the tested methods, chain-of-thought and role-based prompting produced more detailed and context-aware outputs, while zero-shot prompting offered faster and simpler summaries. Therefore, selecting the correct prompting strategy is essential for achieving high-quality AI-generated summaries in education, research, and professional applications.